ADVANCED MACHINE LEARNING · WORKSHEET 03

Simple Linear Regression Using the OLS Method

from scatter plots to the best-fit line — the complete OLS derivation

Name	Branch / Year	Date
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SECTION 1 THE STORYLINE — FROM DATA TO A PREDICTION QUESTION

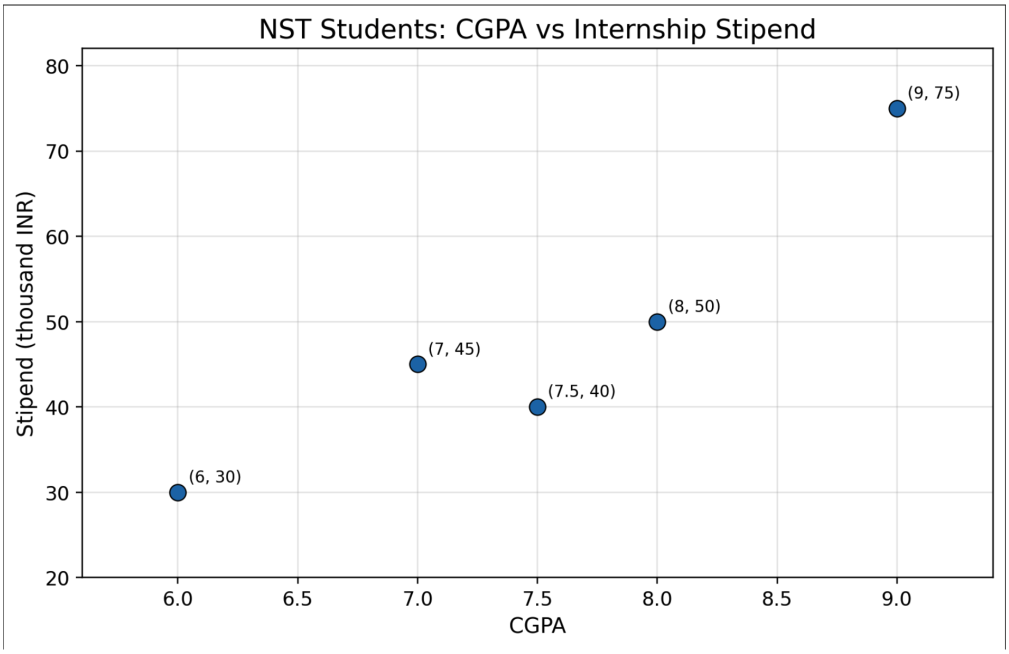
HOOK

Suppose I have a dataset of students at NST. I know their CGPA and the internship stipend they received. If a new student tells me their CGPA, can I predict the stipend they might get?

This is the prediction problem we want to solve. Let's start with real data:

Student	CGPA (x)	Stipend in ₹k (y)
A	8.0	50
B	9.0	75
C	7.0	45
D	7.5	40
E	6.0	30

If you plot this data on a scatter plot (CGPA on x-axis, Stipend on y-axis), you'll observe the points roughly follow a linear pattern — as CGPA increases, stipend tends to increase.



Since the data shows a linear trend, we try to capture this relationship using a **straight-line model**:

$$\hat{y} = m(\text{CGPA}) + c$$

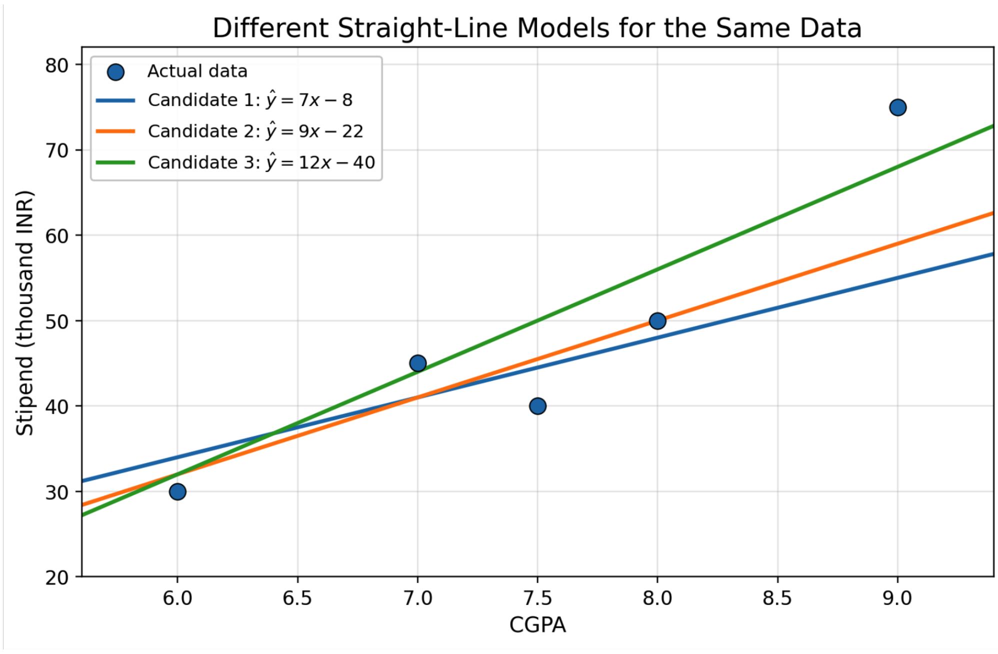
- $\hat{y}$ (y-hat) = the predicted output (predicted stipend)
- x = the input variable (CGPA of the student)
- m = the slope of the line (how steeply stipend changes with CGPA)
- c = the intercept of the line (predicted stipend when CGPA = 0)

In this dataset, we already know the input values (CGPA) and the corresponding output values (stipend). We are learning from examples where the correct answers are given. Therefore, this is an example of **supervised learning**.

SECTION 2 INFINITE CANDIDATE MODELS — WHICH LINE IS BEST?

Now here's the critical realization. The model $\hat{y} = m(\text{CGPA})+c$ has ________ adjustable numbers: ______ and ______. Different students might choose different values:

	Student	Chosen m	Chosen c	Model
Yash	Student 1	10	-30	$\hat{y} = 10x - 30$
Animesh	Student 2	12	-40	$\hat{y} = 12x - 40$
Satvik	Student 3	8	-10	$\hat{y} = 8x - 10$



All three are straight-line models with the same structure, but they use **different values of m and c**.

- **Changing m** changes the slope or tilt of the line.
- **Changing c** shifts the line upward or downward.
- By changing these two numbers, we can create **infinitely many** different straight lines.

KEY INSIGHT

Engineering Problem: Out of infinitely many possible straight lines, which one should I choose? We need an objective, mathematical criterion — not a visual guess.

SECTION 3 WHAT IF I HAVE TWO INPUT VARIABLES?

Before we solve the "which line?" problem, let's briefly extend the example. Suppose along with CGPA, we also have the **IQ score** of each student:

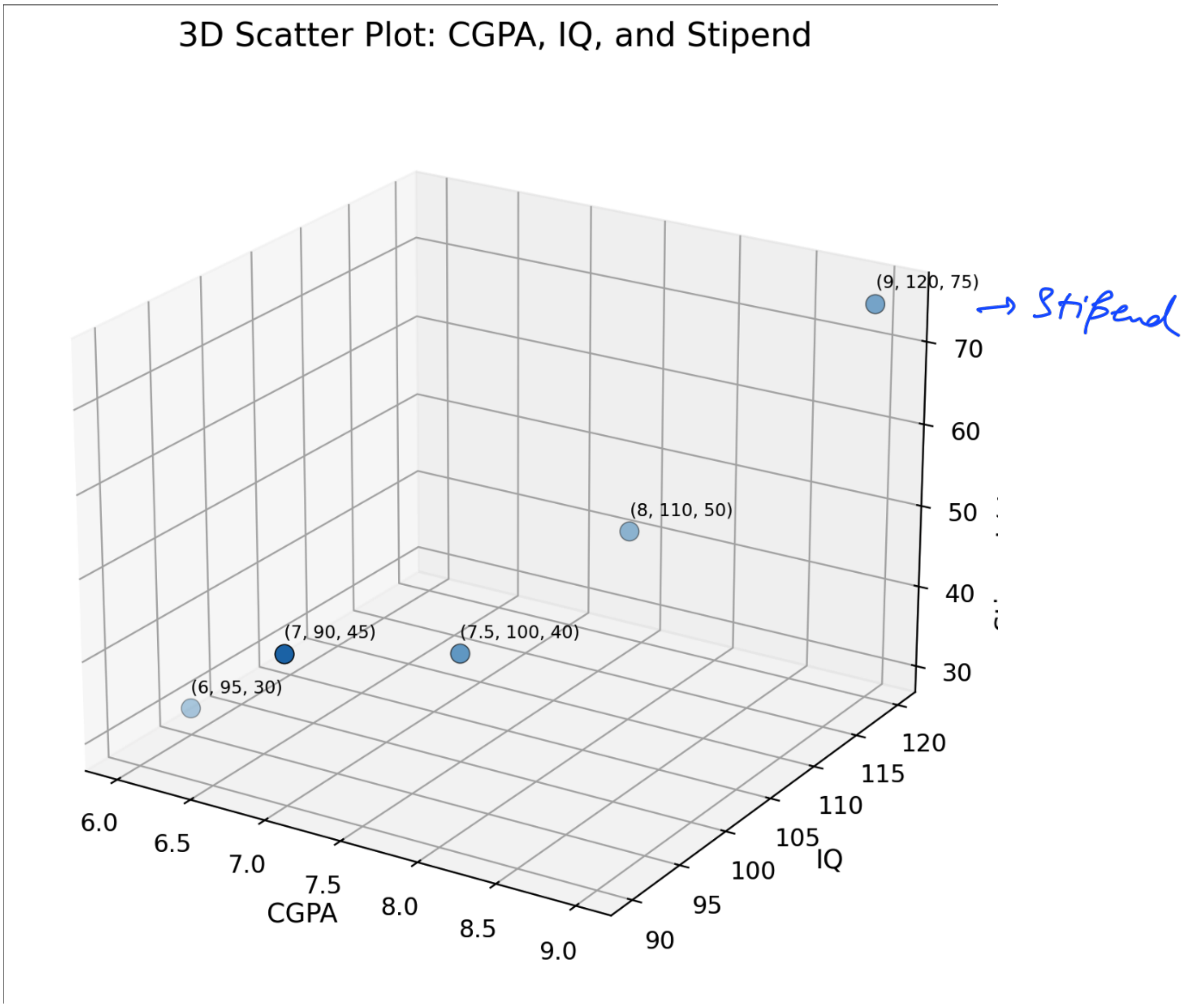
IQ	CGPA	Stipend (₹k)
110	8.0	50
120	9.0	75
90	7.0	45
100	7.5	40
95	6.0	30

With **two input features**, the model becomes:

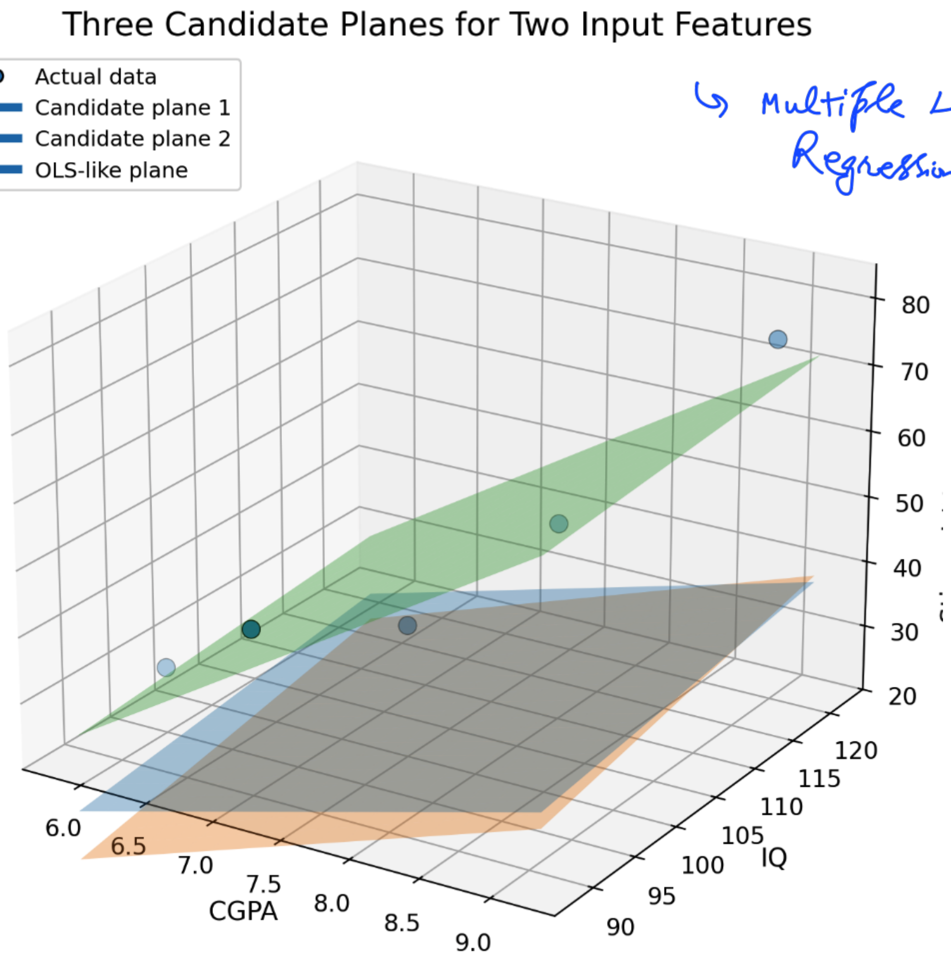
$$\boxed{\hat{y} = m_1(\text{IQ}) + m_2(\text{CGPA}) + c}$$

- $x_1 = \text{IQ}$, $x_2 = \text{CGPA}$
- m_1 tells how prediction changes with IQ
- m_2 tells how prediction changes with CGPA
- c is the intercept term

This is the equation of a **plane in 3D space** (not a line). Just like infinitely many lines existed in the 1-input case, infinitely many planes exist here.



We face the same problem: which plane should we select? The same ideas apply, but with more variables. In this lecture, we keep the discussion focused on **one input variable** so the mathematics remains simple and intuitive.



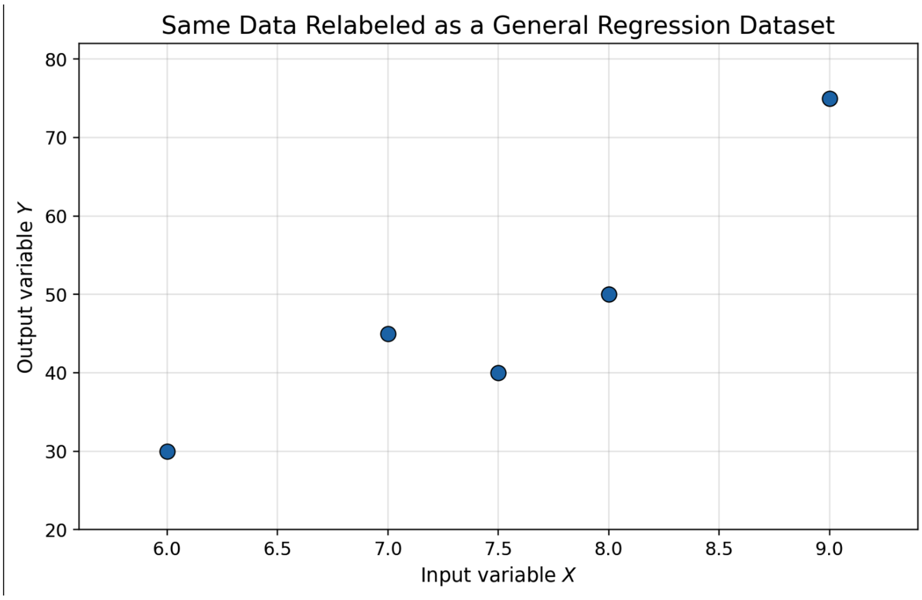
↳ Multiple Linear Regression

> ▷ **PRACTICE P2 · MODEL IDENTIFICATION**
>
> **(a)** In $\hat{y} = m_1(\text{IQ}) + m_2(\text{CGPA}) + c$, how many learnable parameters are there?
> Answer: 3 (List them: m_1, m_2, c)
>
> **(b)** A model with two input features produces a line in 3D space. [F] [True/False]
>
> **(c)** In the one-input case, the model is a Line. In the two-input case, it becomes a Plane. [Fill]

SECTION 4 FORMAL NOTATION — RESIDUALS AND ERRORS

Returning to one variable. Instead of saying "stipend" and "CGPA" every time, we use **general notation**:

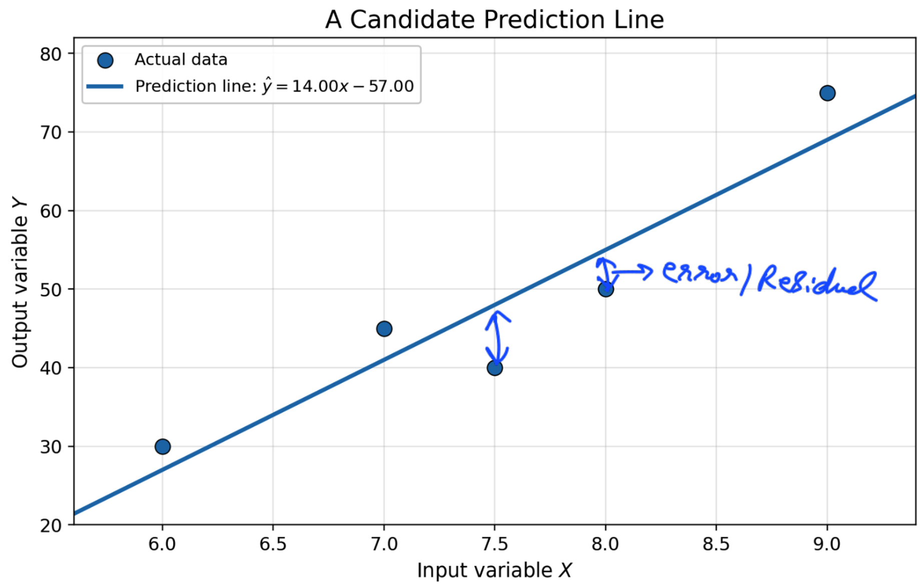
- y = actual output value from the dataset
- x = input value from the dataset
- Each data point is written as: (x_i, y_i) where $i = 1, 2, \ldots, n$



For a given straight-line model, the predicted output for the i-th point is:

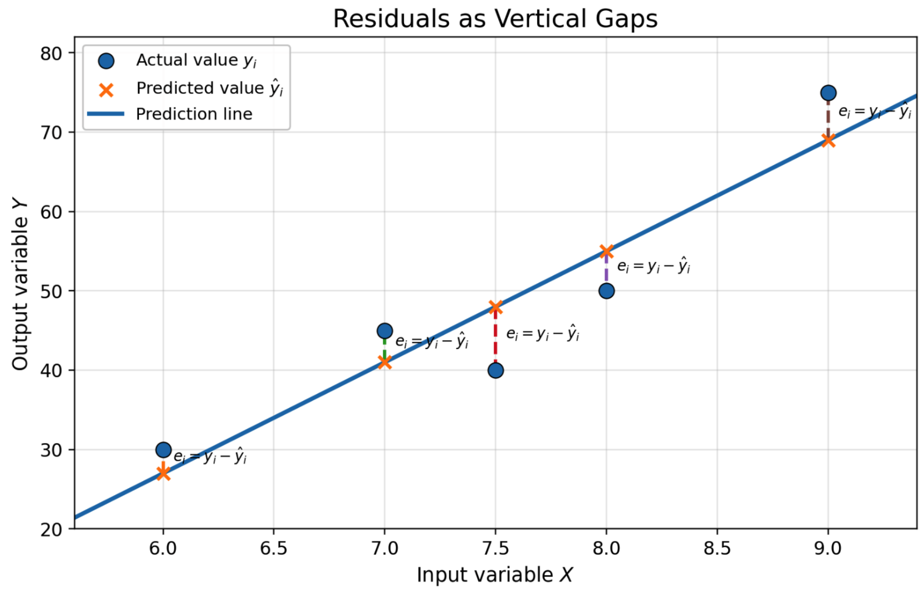
$$\boxed{\hat{y}_i = mx_i + c}$$

y_i'



The model line will usually NOT pass exactly through all data points. The mismatch between the actual value and the predicted value is called the **error or residual** at that point:

$$\boxed{e_i = y_i - \hat{y}_i}$$



Substituting the model equation:

Total Residual

$$\boxed{e_i = y_i - (mx_i + c)}$$

When drawn on a graph, these errors appear as **vertical gaps** from each data point to the prediction line. We measure vertical distances because, in regression, x is treated as fixed and we predict only the corresponding y.

KEY INSIGHT

If the errors are small, the model is good. If the errors are large, the model is bad. So to choose the best model, we need to choose values of m and c that make the errors as small as possible.

▷ PRACTICE P3 · COMPUTE A RESIDUAL

Suppose the model is $\hat{y} = 10x - 30$ and a data point is $(x = 8.0, y = 50)$.

Step 1 — Compute predicted value:

$\hat{y} = 10 \times$ 8 $- 30 =$ 50

(8, 60) ⇒ $\hat{y} = 50$

$y = 60$

Step 2 — Compute the residual:

$e = y - \hat{y} =$ 50 $-$ 50 $=$ 0

$e = 10$

(b) Is this residual positive or negative? What does the sign mean?

SECTION 5 BUILDING THE LOSS FUNCTION — THREE ATTEMPTS

We know we want to minimize errors. But how do we combine all individual errors into a single number that measures overall model quality? This single number is called a **loss function**.

Attempt 1: Minimize the Total Signed Error

$\boxed{TE = 0}$

A first natural idea — just add all the errors:

$$\boxed{\text{Total Signed Error} = \sum_{i=1}^{n} e_i = \sum_{i=1}^{n} (y_i - \hat{y}_i)}$$

This seems reasonable: the model with the smallest total error should be the best. But there is a fatal problem.

Positive and negative errors cancel each other! Consider these six errors:

Point	1	2	3	4	5	6
Error e_i	$+10$	$+5$	$+4$	-10	-3	-6

▷ PRACTICE P4 · THE CANCELLATION TRAP

Using the errors: $+10, +5, +4, -10, -3, -6$

(a) Compute the total signed error:

$\sum e_i = 10 + 5 + 4 + (-10) + (-3) + (-6) =$ 0

(b) [True/False] A total signed error of 0 means the model is perfect. [F]

(c) Why is this total misleading?

____

△ WARNING THE CANCELLATION PROBLEM

The plain sum of errors is NOT a reliable loss function. A total of zero does not mean the model is good — it only means positive and negative errors cancelled each other.

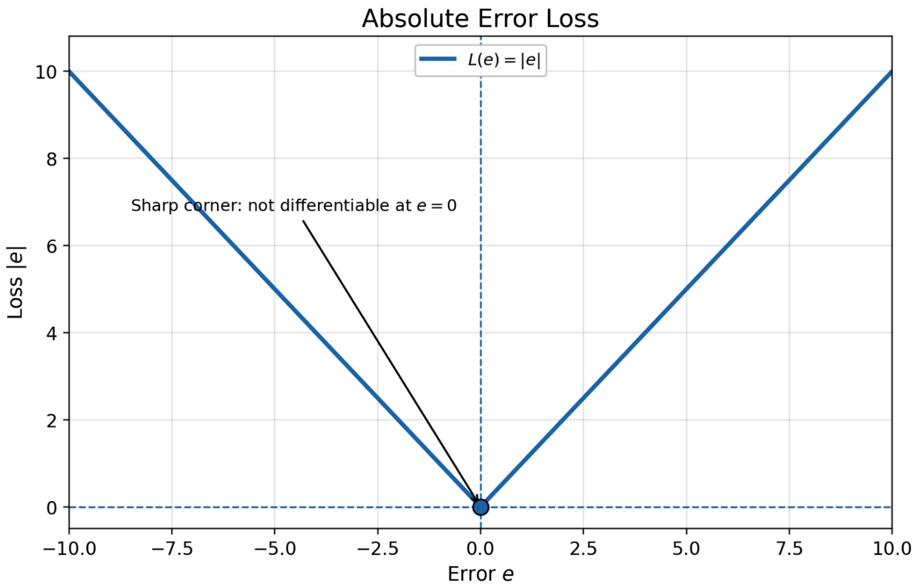
Attempt 2: Use Absolute Error

Remove the sign by taking the absolute value:

$$\text{Total Absolute Error} = \sum_{i=1}^{n} |e_i| = \sum_{i=1}^{n} |y_i - mx_i - c|$$

This solves the cancellation problem — whether an error is positive or negative, $|e|$ is always non-negative. Errors above the line and below the line can no longer cancel.

However, for the OLS method, absolute error creates a mathematical difficulty:



- The graph of $|e|$ has a sharp V-shape at $e = 0$.
- For $e < 0$:

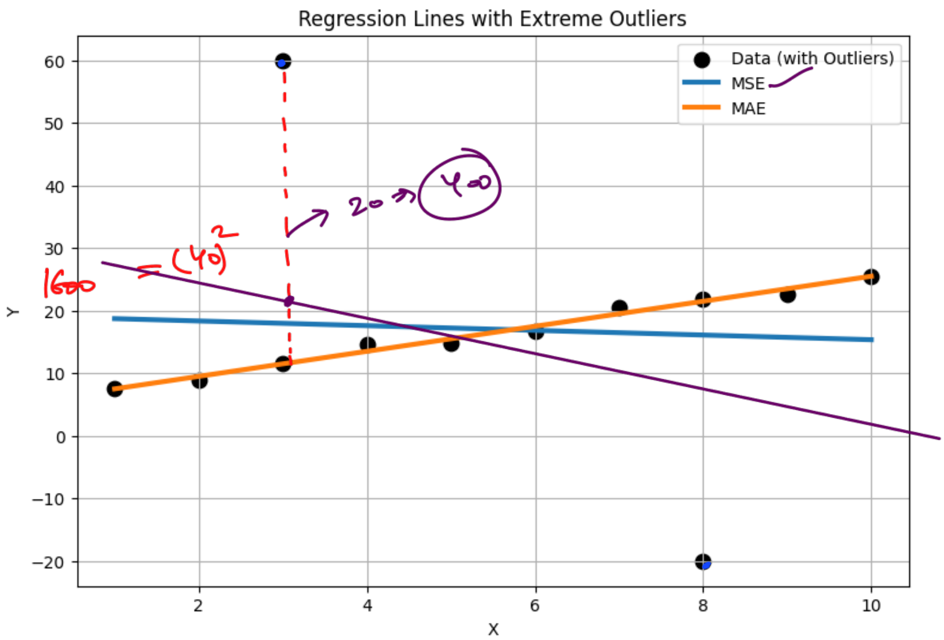
$$\frac{d}{de}|e| = -1$$

  For $e > 0$:

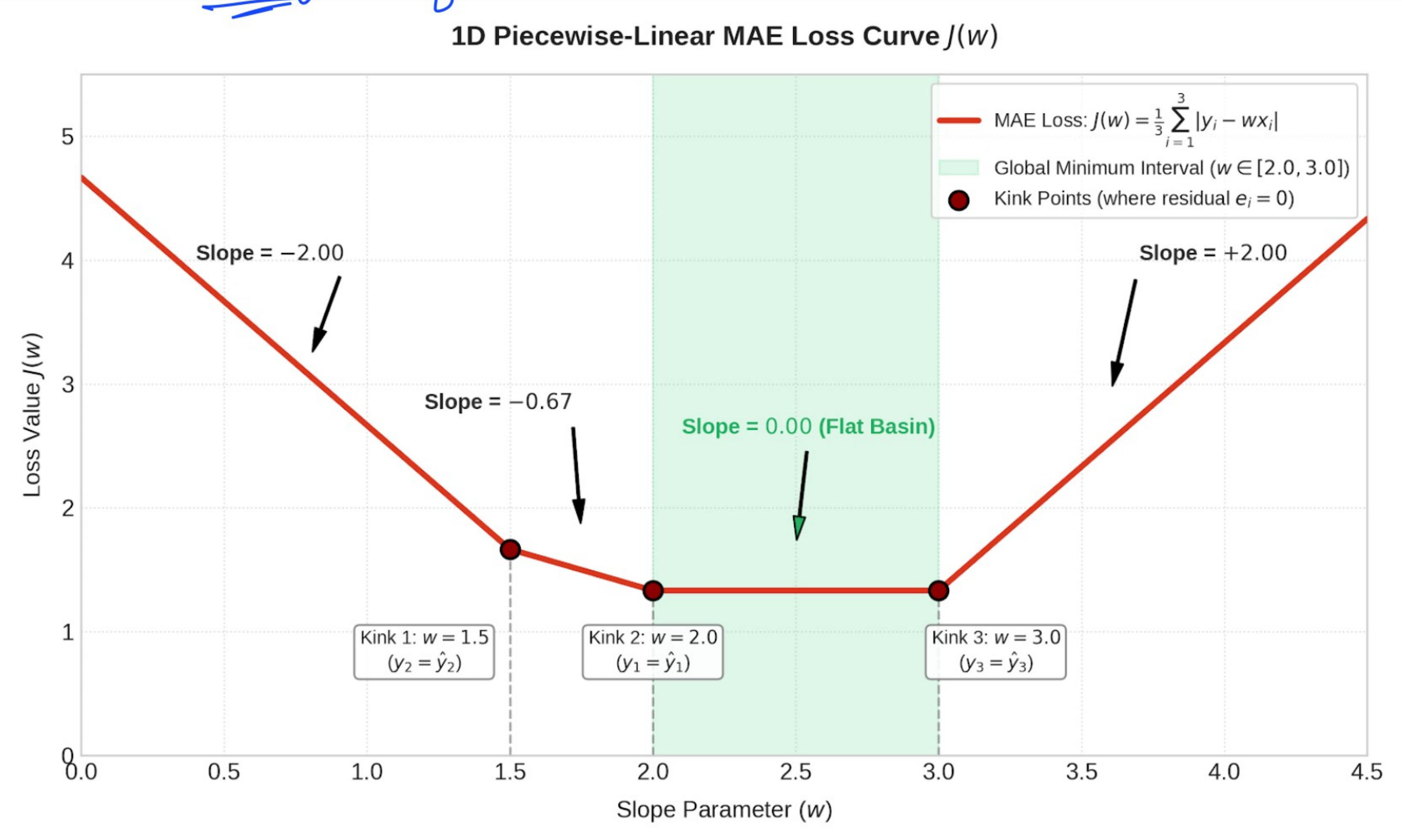
$$\frac{d}{de}|e| = +1$$

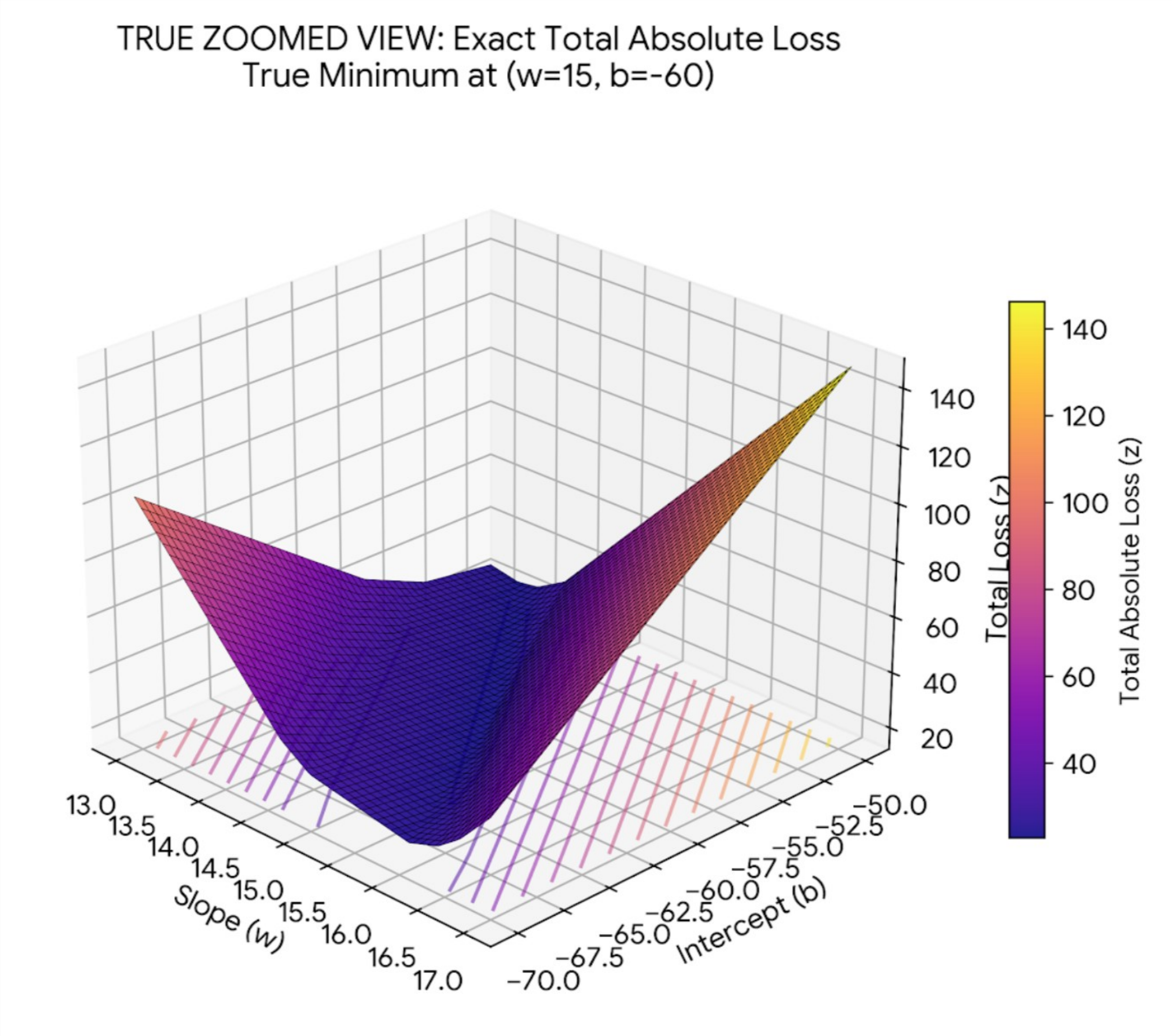
  The derivative is $+1$ for $e > 0$ and -1 for $e < 0$.
- But at $e = 0$, the derivative is undefined because the graph has a sharp corner.
- This means $|e|$ is not smoothly differentiable everywhere — which blocks the clean OLS derivation.

Why absolute error is robust to outliers:
Absolute error penalizes errors linearly. If the error doubles, the penalty also doubles. An error of 1 gets penalty 1. An error of 10 gets penalty 10. The large error is penalized more, but not disproportionately. This is why a single outlier cannot dominate the total loss.



Linear Programming





Sensitive to outliers

Attempt 3: Use Squared Error ✓

Square each error, then add:

$$E(m,c) = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - mx_i - c)^2$$

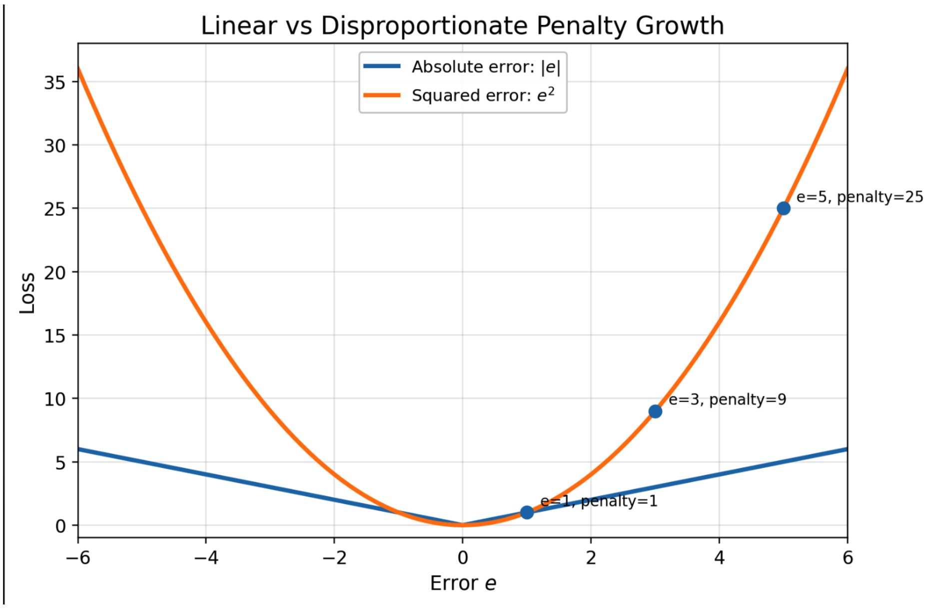
This is the loss function used in **Ordinary Least Squares**. It has three important properties:

- **Always non-negative:** Squaring removes the sign. No cancellation possible.
- **Differentiable everywhere:** The function x^2 is a smooth parabola. Its derivative is $2x$ — defined at every point, including $x = 0$.
- **Penalizes large errors quadratically:** A large error gets a much larger penalty than a small error.

Why squared error is sensitive to outliers:
If the error doubles from 1 to 2, the squared penalty goes from 1 to 4 (four times larger). If the error is 10, the squared penalty is 100. So an error that is 10× larger receives a penalty that is 100× larger. This is why outlier points can strongly influence the OLS best-fit line.

Comparison Table: Absolute vs. Squared Loss

Error (e)	$\lvert e\rvert$ **(Absolute)**	e^2 **(Squared)**
1	1	1
2	2	4
10	10	100



▷ **PRACTICE P5 · LOSS FUNCTION SHOWDOWN**

(a) Complete the table:
Error = 3: $|e|$ = 3 e^2 = 9
Error = 5: $|e|$ = 5 e^2 = 25
Error = 20: $|e|$ = 20 e^2 = 400
(b) [True/False] Signed error can give a total of 0 even when every individual error is nonzero. [T]
(c) [True/False] Absolute error $|e|$ is differentiable at $e = 0$. [F]

MSE

D2

Non convex function

L

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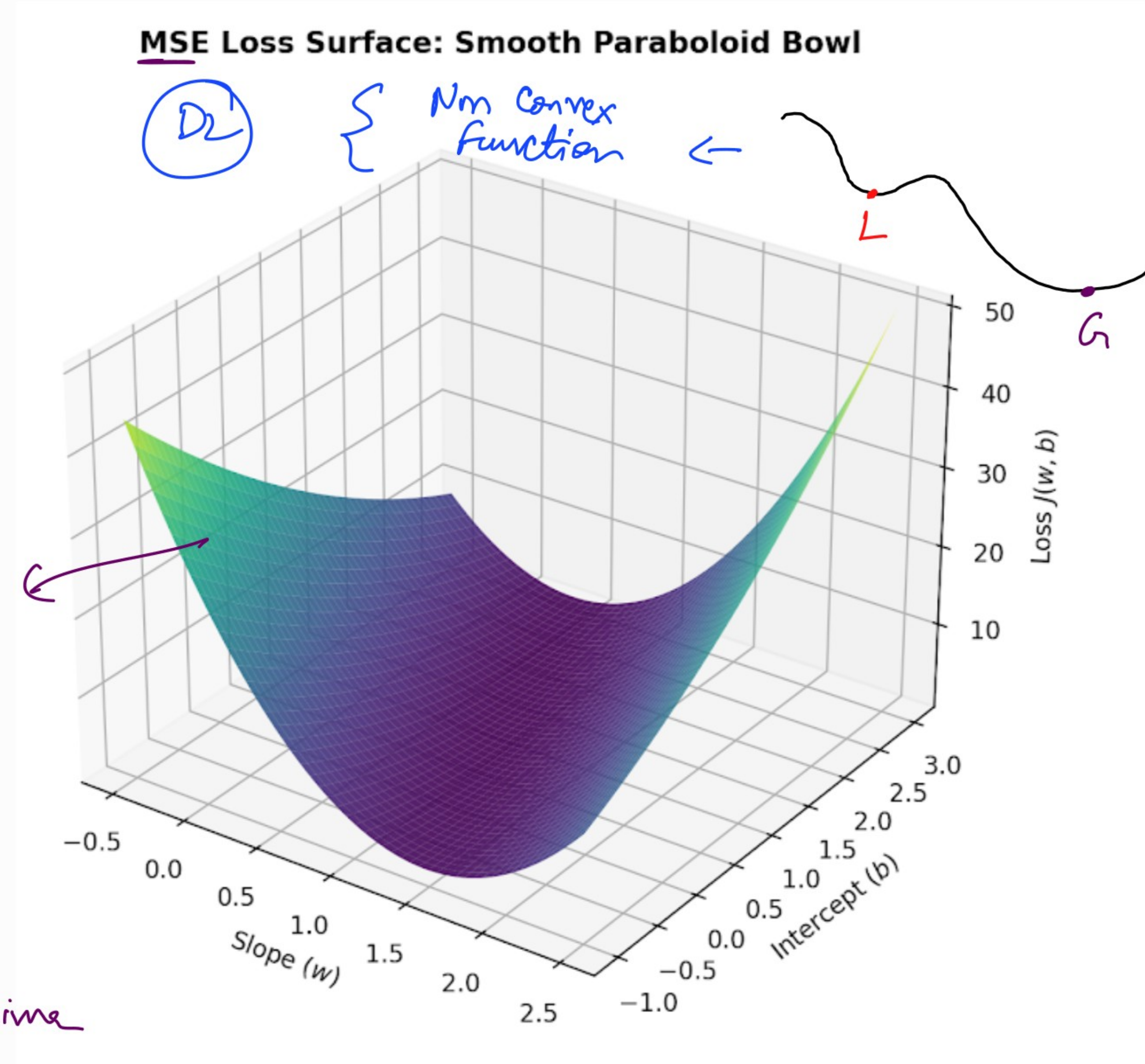
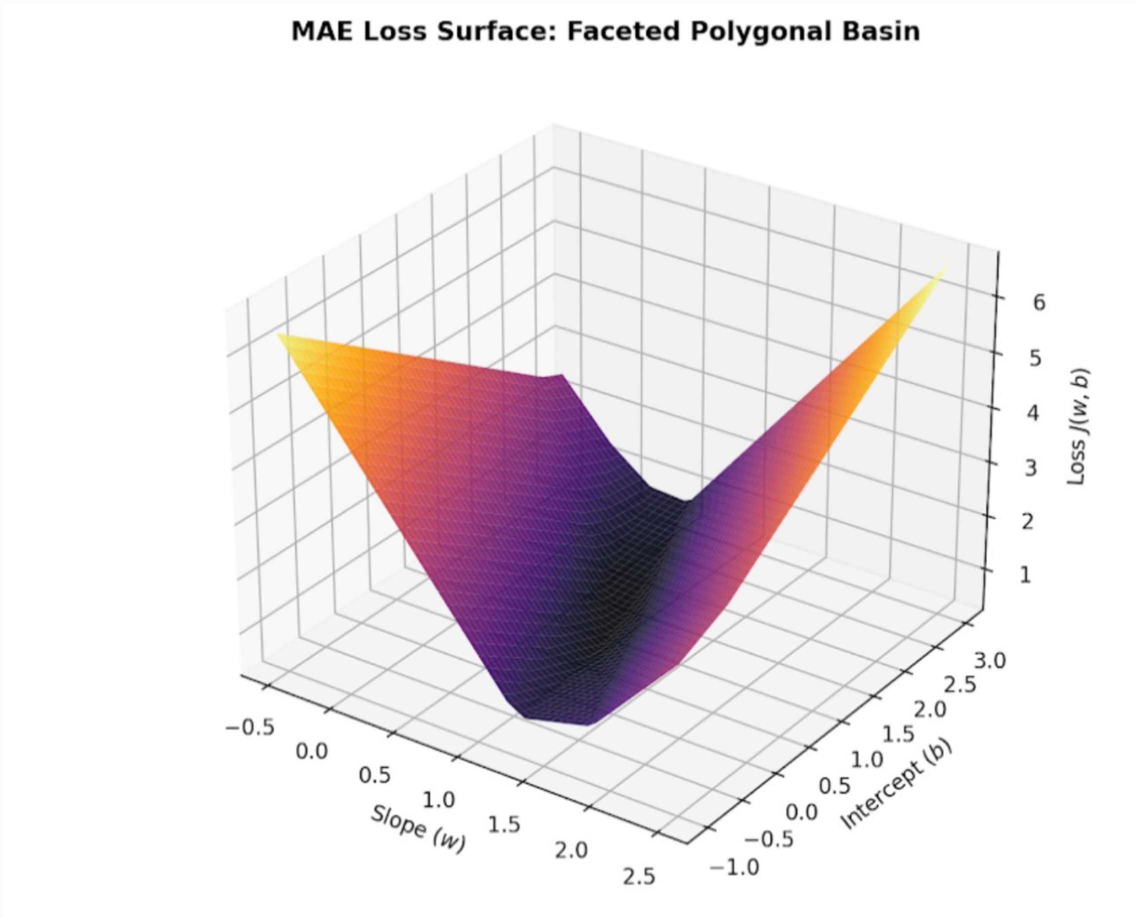


Convex function

↓

Local Minima is same as Global Minima



100 4 1

(d) [True/False] Squared error penalizes an error of 10 exactly 10 times more than an error of 1.
[F]
(e) Match each property (1–3) to the loss function (A–C):
1. Cancellation problem → [C] A. Squared Error (e^2)
2. Smooth and differentiable → [A] B. Absolute Error ($|e|$)
3. Robust to outliers → [B] C. Signed Error ($\sum e_i$)

TAKEAWAY

We choose squared error because it is non-negative (no cancellation), differentiable everywhere (clean calculus), and strongly penalizes large errors. The trade-off: sensitivity to outliers.

SECTION 6 CLOSED-FORM VS. ITERATIVE SOLUTIONS

Before minimizing $E(m, c)$, we need to understand two ways of solving optimization problems in ML:

Approach	What It Means	Example
Closed-Form Solution	An exact formula for the answer. Directly compute optimal parameters.	OLS for Simple Linear Regression (this lecture)
Iterative Solution	Start with initial guesses. Update step-by-step until the error is small.	Gradient Descent (future lectures)

In this lecture, we focus only on the OLS closed-form solution. We will derive exact formulas for m and c using calculus.

▷ **PRACTICE P6 · MATCH THE APPROACH**

Match each description (1–3) to the approach (A–B):
1. "I have a direct formula; I plug in data and get the answer." → [A]
2. "I start with random values and improve them over many rounds." → [B]
3. "OLS for linear regression" → [A]
A. Closed-Form Solution B. Iterative Solution

SECTION 7 VISUALIZING THE ERROR SURFACE

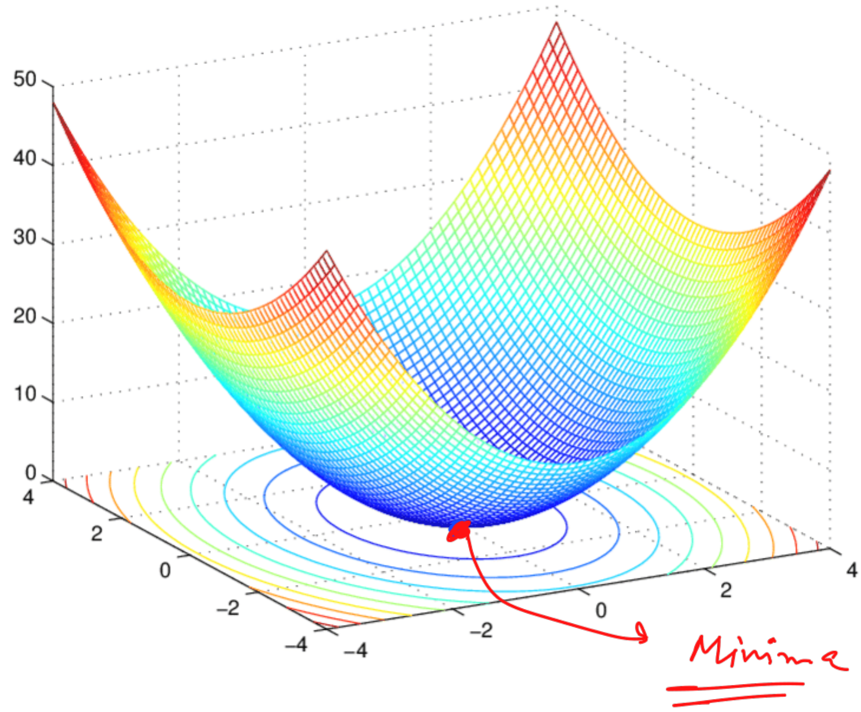
Our squared error function is:

$$E(m, c) = \sum_{i=1}^{n} (y_i - mx_i - c)^2$$

For every pair of values (m, c), we can calculate one value of E. Imagine a 3D surface:

- x-axis represents m (slope)
- y-axis represents c (intercept)
- z-axis represents $E(m, c)$ (the total squared error)

For simple linear regression, this surface is a **convex, bowl-shaped surface**. The bottom of the bowl is the global minimum — the best (m, c) pair.



At the minimum, the surface is **flat in both directions**. In calculus:

$$\frac{\partial E}{\partial c} = 0 \quad \text{and} \quad \frac{\partial E}{\partial m} = 0$$

Because the surface is convex, this stationary point is guaranteed to be the **global minimum** (not just a local one). Now we derive the formulas.

SECTION 8 DERIVING THE INTERCEPT C (EQUATION 1)

We start from:

$$E(m,c) = \sum_{i=1}^{n} (y_i - mx_i - c)^2$$

Step 1: Take the **partial derivative with respect to c** and set it to zero.

$$\frac{\partial E}{\partial c} = 0$$

Step 2: Apply the **chain rule**.

$$\frac{\partial E}{\partial c} = \frac{\partial}{\partial c}\left[\sum_{i=1}^{n} (y_i - mx_i - c)^2\right]$$

$$\frac{\partial E}{\partial c} = \sum_{i=1}^{n} \frac{\partial}{\partial c} (y_i - mx_i - c)^2 \rightsquigarrow \sum_{i=1}^{n} \left(\frac{\partial}{\partial c} r_i^2 \right)$$

Let

$$\boxed{r_i = y_i - mx_i - c}$$

Then

$$\frac{\partial}{\partial c} r_i^2 = \left(\frac{\partial r_i^2}{\partial r_i}\right) \cdot \left(\frac{\partial r_i}{\partial c}\right) = 2r_i \left(\frac{\partial r_i}{\partial c}\right)$$

$$\frac{\partial r_i}{\partial c} = -1$$

and therefore

$$\frac{\partial}{\partial c}(y_i - mx_i - c)^2 = 2(y_i - mx_i - c)(-1)$$

Since y_i and mx_i are constants with respect to c, the derivative of $-c$ with respect to c is -1.

$$\frac{\partial E}{\partial c} = \sum_{i=1}^{n}(-2)(y_i - mx_i - c) = 0$$

Step 3: Set equal to zero and divide both sides by -2:

$$\boxed{\sum_{i=1}^{n}(y_i - mx_i - c) = 0}$$

Step 4: Separate the summation:

$$\sum_{i=1}^{n} y_i \quad - m\sum_{i=1}^{n} x_i \quad - \sum_{i=1}^{n} c \quad = 0$$

$$\sum_{i=1}^{n} y_i \quad - m\sum_{i=1}^{n} x_i \quad - \quad nc = 0$$

Why does $\sum c$ become nc?

Fill in: ____________________

Hint: c is a constant. Adding c to itself n times gives...

Step 5:

$$nc = \sum_{i=1}^{n} y_i - m\sum_{i=1}^{n} x_i \Rightarrow c = \boxed{\frac{1}{n}\sum_{i=1}^{n} y_i - m\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right)}$$

$$c = \bar{y} - m(\bar{x})$$

Recognize the sample means:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

Substituting:

$$\bar{y} - m\bar{x} - c = 0$$

Step 6: Solve for c:

$$\boxed{c = \bar{y} - m\bar{x}} \quad \text{(Equation 1)}$$

KEY INSIGHT

Beautiful intuition: The best-fit OLS line always passes through the point $(\bar{x}, \bar{y})$ — the center (mean) of the data!

▷ PRACTICE P7 · DERIVATION FILL-IN

(a) The derivation contains one mathematical mistake. Identify it and write the correct step.

$$\sum_{i=1}^{n}(y_i - mx_i - c) = 0 \Rightarrow \sum_{i=1}^{n} y_i - m\sum_{i=1}^{n} x_i - \sum_{i=1}^{n} c = 0 \Rightarrow \sum_{i=1}^{n} y_i - m\sum_{i=1}^{n} x_i - c = 0$$

____

(b) The intercept depends only on the mean of the outputs. True /False [ ]

(c) The equation $\sum c = c$ is correct. True /False [ ]

(d) What does the equation $c = \bar{y} - m\bar{x}$ Tell us about where the regression line passes?

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SECTION 9 DERIVING THE SLOPE M (EQUATION 2)

Now substitute $c = \bar{y} - m\bar{x}$ back into the error function:

$$E(m, c) = \sum_{i=1}^{n} (\; y_i - m x_i - c \;)^2$$

$$c = \bar{y} - m\bar{x}$$

$$E(m) = \sum_{i=1}^{n} (\; y_i - m x_i - \bar{y} + m\bar{x} \;)^2$$

Simplify inside the bracket:

$$E(m) = \sum_{i=1}^{n} (\; \underbrace{(y_i - \bar{y})}_{a_i} \; - m \underbrace{(x_i - \bar{x})}_{b_i} \;)^2$$

Now define **shorthand** for clarity:

- Let $\cancel{x_i}\, b_i = x_i - \bar{x}$ (deviation of each x from the mean)
- Let $\cancel{y_i}\, a_i = y_i - \bar{y}$ (deviation of each y from the mean)

Then:

$$\boxed{E(m) = \sum_{i=1}^{n} (a_i - m b_i)^2}$$

Differentiate with respect to m and set to zero:

$$\frac{dE}{dm} = \sum_{i=1}^{n} 2(a_i - m b_i) \frac{d}{dm}(\; a_i - m b_i \;)$$

$$\frac{d}{dm}(a_i - m b_i) = -b_i$$

$$\frac{dE}{dm} = \sum_{i=1}^{n} 2\,(\; a_i - m b_i \;)(\; -b_i \;) = \boxed{-2 \sum_{i=1}^{n} b_i (a_i - m b_i)}$$

For the minimum, we set this equal to zero:

$$-2 \sum_{i=1}^{n} b_i (a_i - m b_i) = 0$$

Dividing by -2 gives:

$$\boxed{\sum_{i=1}^{n} b_i (a_i - m b_i) = 0}$$

Now we expand the summation:

$$\sum_{i=1}^{n} b_i a_i - \sum_{i=1}^{n} m b_i^2 = 0$$

Since m is constant with respect to i:

$$\sum_{i=1}^{n} b_i a_i - m \sum_{i=1}^{n} b_i^2 = 0$$

Now we solve for m:

$$\boxed{m\sum_{i=1}^{n} b_i^2 = \sum_{i=1}^{n} b_i a_i,} \qquad \boxed{m = \frac{\sum_{i=1}^{n} b_i a_i}{\sum_{i=1}^{n} b_i^2}}$$

Now we substitute back:

$$b_i = x_i - \bar{x}$$

and

$$a_i = y_i - \bar{y}$$

So:

$$m = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\frac{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2} \rightarrow \frac{Cov(X,Y)}{Var(X)} = m$$

Best Fit Line

$$\hat{y} = mx + c$$

I will call this **Equation (2)**:

$$\boxed{m = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}} \qquad \text{(Equation 2)}$$

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$$c = \bar{y} - m\bar{x}$$

KEY INSIGHT

The numerator measures how x and y vary together (covariance). The denominator measures how much x varies by itself (variance). The slope tells how much y changes with x, adjusted for the spread in x.

This formula assumes the **denominator is not zero**. If every student had exactly the same CGPA, there would be no variation and no meaningful slope.

▷ PRACTICE P8 · DERIVATION FILL-IN

(a) Explain in one sentence why the intercept c is substituted before optimizing the slope.

(b) Compute

$$\frac{d}{dm}(a_i - mb_i)^2$$

(c) Starting from

$$\sum_{i=1}^{n} b_i(a_i - mb_i) = 0$$

derive the expression for m.

(d) What does the denominator

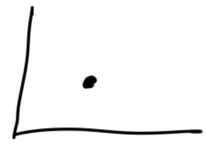
$$\sum_{i=1}^{n}(x_i - \bar{x})^2$$

represent?
A. Covariance B. Variance of x C. Mean of x D. Mean squared error
(e) Why would the slope be undefined if every x_i has the same value?

SECTION 10 THE FINAL OLS MODEL

Now we have both formulas:

$$m = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \qquad c = \bar{y} - m\bar{x}$$



The procedure is:

$\hat{y} = mx + c$

- **Step 1:** Compute m using Equation 2.

$\uparrow (\bar{x}, \bar{y})$

- **Step 2:** Substitute m into Equation 1 to compute c. $\downarrow$
- **Step 3:** Write the final model: $\hat{y} = mx + c$

$\bar{y} = m\bar{x} + c \Rightarrow \cancel{m\bar{x}} + \bar{y} - \cancel{m\bar{x}}$

This is the OLS best-fit line. The phrase "best-fit" means precisely this: it is the line that minimizes the total squared error.

$\bar{y} = \bar{y}$

▷ PRACTICE P9 · ARRANGE THE OLS STEPS

Arrange these steps in the correct order (write 1–6):

[5] Minimize E using partial derivatives ($\partial E/\partial m = 0$ and $\partial E/\partial c = 0$)
[3] Square each residual
[1] Assume the model $\hat{y} = mx + c$
[4] Add all squared residuals to form $E(m, c)$
[2] Calculate the residual e_i at each data point
[6] Obtain the closed-form formulas for m and c

SECTION 11 NUMERICAL PRACTICE — OLS BY HAND

Use a different dataset to test your understanding. You are given 5 students' Study Hours vs Quiz Scores:

Student	Study Hours (x)	Quiz Score (y)
P	1	25
Q	3	40
R	5	55
S	7	65
T	9	80

$\bar{x} = 5$ $\bar{y} = 53$

Reflect 1 · The OLS line passes through $(\bar{x}, \bar{y})$. Verify: does $\hat{y}$ at $x = \bar{x}$ equal $\bar{y}$? Show the calculation.

Hint: $\hat{y} = m \cdot \bar{x} + c$. Substitute $c = \bar{y} - m\bar{x}$.

Reflect 2 · If every student had the exact same number of Study Hours (say, all 5), what would happen to the denominator of the m formula? What does this mean for the model?

Hint: What is $\sum(x_i - \bar{x})^2$ when all x_i are equal?

SECTION 12 CONCEPTUAL WRAP-UP — WHY OLS?

▷ PRACTICE P10 · COMPUTE OLS BY HAND

Step 1 — Compute the means:

$$\bar{x} = \underline{\;5\;\;\;\;\;\;} \qquad \bar{y} = \underline{\;53\;\;\;\;\;}$$

Step 2 — Compute deviations and products

Student P: $x_P = 1 \Rightarrow a_P = \underline{-4} \Rightarrow y_P = 25 \Rightarrow b_P = \underline{-28} \Rightarrow (x_P - \bar{x})(y_P - \bar{y}) = \underline{112} \Rightarrow (x_P - \bar{x})^2 = \_\_\_\_$

Student Q: $x_Q = 3 \Rightarrow a_Q = \underline{-2} \Rightarrow y_Q = 40 \Rightarrow b_Q = \underline{-13} \Rightarrow (x_Q - \bar{x})(y_Q - \bar{y}) = \underline{26} \Rightarrow (x_Q - \bar{x})^2 = \_\_\_\_$

Student R: $x_R = 5 \Rightarrow a_R = \_\_\_\_ \Rightarrow y_R = 55 \Rightarrow b_R = \_\_\_\_ \Rightarrow (x_R - \bar{x})(y_R - \bar{y}) = \_\_\_\_ \Rightarrow (x_R - \bar{x})^2 = \_\_\_\_$

Student S: $x_S = 7 \Rightarrow a_S = \_\_\_\_ \Rightarrow y_S = 65 \Rightarrow b_S = \_\_\_\_ \Rightarrow (x_S - \bar{x})(y_S - \bar{y}) = \_\_\_\_ \Rightarrow (x_S - \bar{x})^2 = \_\_\_\_$

Student T: $x_T = 9 \Rightarrow a_T = \_\_\_\_ \Rightarrow y_T = 80 \Rightarrow b_T = \_\_\_\_ \Rightarrow (x_T - \bar{x})(y_T - \bar{y}) = \_\_\_\_ \Rightarrow (x_T - \bar{x})^2 = \_\_\_\_$

Step 3 — Sum and compute m

$$\sum_{i=1}^{5}(x_i - \bar{x})(y_i - \bar{y}) = \underline{112} + \underline{26} + \_\_\_\_ + \_\_\_\_ + \_\_\_\_ = \_\_\_\_\_\_$$

$$\sum_{i=1}^{5}(x_i - \bar{x})^2 = \underline{16} + \underline{4} + \_\_\_\_ + \_\_\_\_ + \_\_\_\_ = \_\_\_\_\_\_$$

$$m = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{\_\_\_\_\_\_}{\_\_\_\_\_\_} = \_\_\_\_\_\_$$

Step 4 Compute c

$$c = \bar{y} - m\bar{x} = \_\_\_\_\_\_ - \_\_\_\_\_\_ \times \_\_\_\_\_\_ = \_\_\_\_\_\_$$

Step 5 — Final model

$$\hat{y} = \_\_\_\_\_\_\_\_ x + \_\_\_\_\_\_\_\_$$

Step 6 Prediction

$$\hat{y} = \_\_\_\_\_\_ \times 6 + \_\_\_\_\_\_ = \_\_\_\_\_\_$$

▷ PRACTICE P11 · TRUE/FALSE RAPID FIRE

Mark each statement True or False:

(a) The OLS line always passes through every single data point. [F]
(b) The OLS line always passes through $(\bar{x}, \bar{y})$. [T]
(c) Squared error is more robust to outliers than absolute error. [F]
(d) The derivative of $|x|$ exists at $x = 0$. [F]
(e) In $\hat{y} = mx + c$, m and c are the learnable parameters. [T]
(f) OLS gives an iterative (step-by-step) solution. [F]
(g) If all x_i values are identical, the slope m is undefined. [T]
(h) The error surface $E(m, c)$ is convex for SLR. [T]

▷ PRACTICE P12 · OUTLIER IMPACT ANALYSIS

You add a 6th student to the dataset: $(x = 10, y = 20)$. This student studied 10 hours but scored only 20 — a clear outlier.

(a) Under absolute error, the penalty for this point is: $|e| = |20 - \hat{y}| = |20 - (m \times 10 + c)|$. This grows ________________ with the error.

(b) Under squared error, the penalty for this point is: $e^2 = (20-\hat{y})^2$. This grows ________________ with the error.

(c) Which loss function will be more affected by this outlier? Why?

____

(d) Will the OLS best-fit line shift toward this outlier or away from it?

____

Forward Handoff:

In the next lecture, we extend the same idea from one input feature to multiple input features (Multiple Linear Regression). The OLS derivation will use the same principle — minimize squared error — but with matrices and vectors instead of simple algebra. The concept of Gradient Descent (iterative optimization) will also be introduced.

TAKEAWAY

Today: Story → Scatter Plot → Line Model → Infinite Lines → Signed Error Fails → Absolute Error Has Issues → Squared Error Wins → OLS Derivation → Final Best-Fit Model.

BY THE END OF THIS SHEET YOU CAN

- ☐ Represent a real supervised prediction problem as $\hat{y} = mx + c$ and identify all components $(x, y, \hat{y}, m, c)$.
- ☐ Justify why squared-error loss is chosen over signed error and absolute error for OLS.
- ☐ Explain how absolute error is robust to outliers and why squared error is sensitive to outliers.
- ☐ Distinguish between closed-form solutions and iterative solutions in machine learning.
- ☐ Derive the OLS closed-form formulas for the intercept c and slope m from scratch using partial derivatives.
- ☐ Compute OLS estimates by hand for a small dataset and interpret the results.